

Revised Project Plan

Look-alike Domain Analyzer & Reporting Tool

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1. Project Overview

This document is the revised project plan for Phase 2 of our Lookalike Domain Analyzer and Reporting Tool. It contains a review of Phase 1 deliverables, instructor feedback, and an updated scope and timeline for the current semester.

1.1 What We Are Building

The project is a chrome extension tool that allows users to analyze domains and URLs for lookalikes and phishing risks. The tool serves two audiences:

- End users: User submits a domain or URL through the extension and receives an easy-to-understand report that explains if the domain is suspicious and should be avoided.
- Security analysts: Same submission provides full evidence breakdown covering various threat intelligence, heuristic signals, infrastructure data, WHOIS and registrar information, MX records, blacklist status, and complete risk map that could be used for investigations and reporting.

Both models are powered by the same backend analysis pipeline; the level of detail shows is tailored to each audience.

1.2 Key Components

- Chrome extension - manual URL/domain entry, brand override, JSON export, and lookalike candidate display
- AWS-hosted TypeScript backend - Lambda handlers, API Gateway, CDK infrastructure stack, deployed in ca-central-1
- Deterministic scoring engine - weighted signals across lexical, infrastructure, content, reputation, behavioral, and history categories
- AWS Bedrock LLM – used for feature collection and analysis for evidence contributing to threat score, and reporting tool formatting
- Legitimate lookalike generation – DNS Twister based generation of existing potential lookalikes based on a submitted legitimate domain
- Free and open intelligence sources - RDAP, crt.sh, Internet Archive CDX API, OpenPhish, and PhishTank
- Optional paid intelligence providers - VirusTotal, AbuseIPDB, Google Safe Browsing (key-gated, graceful degradation when unavailable)

2. Phase 1 Summary – What Was Completed

Phase 1 produced a working proof of concept that established the foundational architecture and analyst workflow. The following capabilities are not fully optimized, but were built and demonstrated at the end of the phase:

- Manual URL/domain analysis
- Brand recognition and override
- Deterministic scoring
- Analyst-facing evidence display
- JSON export
- DNSTwister-based lookalike generation

3. Feedback Received and How It Was Addressed

The following summarizes the key feedback received at the end of Phase 1 and the corresponding adjustments made to Phase 2 scope and approach...

Criticism: Proof of Concept demonstration was simple and does not stand out among similar existing tools.

Response: We will continue developing and improving the functionality and design of the extension to ensure all the planned features are implemented. The focus of phase 2 will be to build features that existing tools do not combine in one place.

Recommendation: Add feature that visits the lookalike website to observe behaviour as an addition to logic for threat score.

Response: We will implement a sandbox environment that will visit the lookalike website to observe behaviours and evaluate the overall look of the website to be included in the determination of the threat score.

Recommendation: Alert users that have extension enabled they may be visiting a potential lookalike or malicious website.

Response: As a part of our Chrome extension, there will be a live monitoring capability that will alert the user of a potential lookalike, recommending they leave the website or provide warning based on the severity of the threat-score.

4. What Still Needs to Be Completed

Phase 2 closes the following gaps identified from Phase 1 and the feedback review:

4.1 Technical

- Optimized accuracy of threat score logic and fine-tuning severity levels of score.
- Recognition of brand as the correct existing entity that is being impersonated.

4.2 Infrastructure

- Finalized AWS environment.

4.3 Design

- A completed Chrome extension that features two different levels of depth, one with more features involved with logic for security analysts and a simpler one for end-users.
- A comprehensive dashboard that showcases the tool's full capability, featuring granular threat score analysis and reporting tool for security analysts.

4.4 Reporting

- A functional reporting tool that exports evidence and information into universal JSON format for time-effective takedown procedures.

5. Final Deliverables

The following artifacts will be produced by the end of Phase 2:

Deliverable	Description
Chrome Extension	Chrome extension built against the deployed production API, with both the basic user view and the full analyst view implemented, and the reporting module integrated
Deployed AWS Backend	Live deployment across dev, test, and prod environments with Secrets Manager, CloudWatch, and DynamoDB lifecycle configured
Basic User Result View	Simplified, plain-language safety result for non-technical users explaining why a domain is flagged and what action to take
Full Analyst Result View	Complete evidence breakdown including all threat intelligence, heuristic signals, infrastructure data, WHOIS/registrar info, MX records, and blacklist status

Human-Readable Analyst Report	Exportable analyst report (structured format) in addition to the existing JSON export, suitable for distribution to registrars, brands, or security teams
Scoring Corpus and Calibration	Labeled sample set used to calibrate and validate scoring weights, documented for reproducibility
Expanded Test Suite	End-to-end analyst workflow tests and degraded-provider condition tests in addition to existing unit and integration coverage
Release Documentation	Documented deployment and release process covering packaging, environment promotion, and smoke-test procedures
Final Demo and Presentation	Live demonstration covering the full Phase 2 workflow: domain submission, basic user view, analyst view, lookalike generation, and report export

6. How We Plan to Complete the Work

We are organizing phase 2 into four workstreams across four milestones throughout the span of the allocated thirteen weeks.

6.1 Workstreams

Workstream	Scope
Technical	Provider hardening, signal completion, redirect-chain analysis, content analysis improvements, phishing-template detection, brand catalog expansion, scoring calibration, test coverage expansion
Infrastructure	AWS environment formalization, Secrets Manager migration, CDK cleanup, DynamoDB lifecycle, CloudWatch deployment, release process documentation, extension packaging against live API
Design & User Experience	Basic user simplified view, automatic tab detection, evidence hierarchy redesign, degraded-state UI improvements, lookalike batch review, layout and typography consistency
Reporting	Registrar and brand contact expansion, CERT/CSIRT routing, Bedrock dependency reduction for core fields, human-readable report format, audience-specific report views

6.2 Milestones

Milestone	Focus	Outcome
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Milestone 1 (Weeks 1–3)	Provider hardening, infrastructure signal completion, passive history expansion, provider diagnostics cleanup	Reliable backend evidence model — all signals backed by real logic or explicit failure states
Milestone 2 (Weeks 4–6)	Content analysis improvements, phishing-template detection, brand-matching expansion, score calibration	Higher-quality detection and defensible, calibrated scoring
Milestone 3 (Weeks 7–9)	Basic user view, analyst UX cleanup, lookalike-review improvements, structured report output, expanded reporting coverage	Both user-facing views complete and a full reporting package
Milestone 4 (Weeks 10–13)	AWS environment maturity, Secrets Manager, CloudWatch, release packaging, expanded tests, deployment validation, final demo	Stable Phase 2 release candidate and submission-ready deliverables

7. Weekly Milestone Breakdown

Weeks 1–3: Foundation and Backend Reliability (Milestone 1)

Week	Workstream(s)	Key Activities
Week 1	Technical	- Review PoC to identify scoring gaps (providers, scoring gaps, etc.) - Optimize timeout and rate limit handling - Research and finalize external intelligence providers (VirusTotal, AbuseIPDB, etc.)
Week 2	Technical	- Complete finalized providers and diagnostic workflows - Implement logic for redirect-chain analysis and final destination resolution - Begin implementing placeholder features from PoC
Week 3	Technical + Infrastructure	- Improve passive DNS and historical ownership intelligence for score logic - Confirm backend evidence pipeline

Weeks 4–6: Detection and Scoring (Milestone 2)

Week	Workstream(s)	Key Activities
Week 4	Technical	• Improve content analysis (text similarity, asset reuse checks, form-action classification)
Week 5	Technical	• Begin working on a stronger phishing template / visual similarity detection • Optimize brand recognition to include less recognizable typo/confusable matching
Week 6	Technical	• Confirm overall detection and scoring accuracy and perform final improvements • Accurately separate real threats from false positives

Weeks 7–9: UX and Reporting (Milestone 3) and Environment (Milestone 4)

Week	Workstream(s)	Key Activities
Week 7	Design + Technical	• Improve extension layout prioritizing an intuitive experience for users • Begin designing and planning reporting tool module implementation
Week 8	Design	• Improve extension usability (errors, loading, layout) • Begin cloud hosting architecture research for design • Continue developing reporting tool module
Week 9	Reporting + Design	• Finalize implementation of reporting module • Adding contacts, reports, proper routing • Optimizing contacts found • Finalize plans for cloud hosting implementation

Weeks 10–13: Final Optimization and Delivery

Week	Workstream(s)	Key Activities
Week 10	Infrastructure	• Implement secret management and cleaning CDK depreciations • Begin implementing finalized plans for cloud hosting
Week 11	Infrastructure + Technical	• Adding DynamoDB lifecycle and caching • Adding CloudWatch dashboard, logs, alerts
Week 12	Infrastructure + Technical	• Test full system • Fix remaining bugs and add final requirements
Week 13	All Workstreams	• Final testing and bug fixes • Preparing final demo, documentation, and submission

8. Risks, Challenges, and Blockers

Risk	Likelihood	Impact	Mitigation
External provider API rate limits or key access delays	Medium	High	Obtain and validate all API keys by end of Week 2. Implement graceful degradation so the tool remains functional when a provider is unavailable.
LLM inference speed vs cost trade off	High	Medium	For phase 2, the current model tier is acceptable for a capstone demo environment. If the latency becomes a blocker, the team will evaluate switching to a more cost-effective hosting platform.